

Mridulla Ganesh

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EDUCATION

Purdue University

Computer Engineering

West Lafayette, IN

August 2023 — May 2027

- Cumulative GPA: 3.4/4.0 | Dean's List, Semester Honors
- Relevant Coursework: Computer Network Systems, Operating Systems, Computer Design, Embedded Systems

WORK EXPERIENCE

New Product Introduction Co-Op

May 2026 — Present

Advanced Micro Devices

Austin, TX

- Designing thermal models for System Level Test (SLT) environments to improve test-flow accuracy for next-generation AMD products
- Extending and scaling existing modeling frameworks to support new product platforms and improve cross-product adaptability
- Driving SLT bring-up efforts for new AMD desktop products through validation, debugging, and test-flow development

Undergraduate Teaching Assistant

August 2025 — May 2026

Purdue University

West Lafayette, IN

- Mentoring undergraduate students in embedded systems programming in C for the RP2350 microcontroller
- Guiding students in writing programs to interface with hardware peripherals, including GPIO, UART, SPI, and timers
- Performing root cause analysis on student projects with logic errors and hardware faults, teaching problem solving at the hardware-software interface

Quality Engineering Intern

May 2025 — August 2025

Coherent

Monroe, CT

- Wrote Python scripts using OpenCV to collect and process image data for an AI defect detection model, helping improve the model's detection accuracy for surface flaws on semiconductor wafer chucks
- Built interactive PowerBI dashboards to visualize quality metrics, providing the engineering team with clearer insights into manufacturing trends
- Assisted with root cause analysis on a critical component failure by performing physical inspections and identifying the cause

RESEARCH EXPERIENCE

Research with VIP SoCET, Datacenter Network Accelerator

August 2025 — Present

Purdue University

West Lafayette, IN

- Developing a software simulation of a server environment, creating a model of a custom hardware network accelerator to enable pre-silicon validation
- Implementing network communication layers to bridge the software simulation with the physical hardware accelerator over Ethernet, creating a testing environment
- Designing and scripting test workloads to validate the accelerator's functionality and benchmark its performance

Undergraduate Research Assistant, AI Network Performance

January 2025 — Present

Purdue University, w/ Dr. Xiaoqi Chen

West Lafayette, IN

- Analyzing modern data center network architecture to model their performance impact on large-scale AI training workloads
- Utilizing network simulators including ns-3 and AstraSim to analyze data flow, identifying key network bottlenecks in collective communication patterns

Undergraduate Research Assistant, GPU Acceleration

January 2025 — May 2025

Purdue University, w/ Dr. Milind Kulkarni

West Lafayette, IN

- Accelerated k-nearest neighbors algorithm by leveraging NVIDIA GPU ray-tracing cores for high-speed spatial queries
- Optimized the core search algorithm in CUDA and C++, using rotations to optimize for data in different Minkowski distances

SKILLS

- Programming Languages:** C, Python, Java, MATLAB, RISC-V Assembly, C++, CUDA
- Technologies:** Linux, Git, ns-3, htsim, System Verilog